

NeRF vs 3D Gaussian Splatting: A Comparative Review

Course: EE60030 — Geometric Methods in Computer Vision (GMCV)

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Mathematical Preliminaries — Camera & Ray Models

Objective: Generate photorealistic images from arbitrary viewpoints from sparse, calibrated 2D images.

Camera Projection Model: Map 3D points

$\mathbf{X} \in \mathbb{R}^3$ to pixels (u, v) via intrinsic $\mathbf{K}$ and extrinsic $[\mathbf{R} \mid \mathbf{t}]$ matrices:

$$\lambda(u, v, 1)^T = \mathbf{K}[\mathbf{R} \mid \mathbf{t}](\mathbf{X}^T, 1)^T$$

Ray Casting (Inverting Projection): Ray fired from camera center $\mathbf{o}$ through pixel:

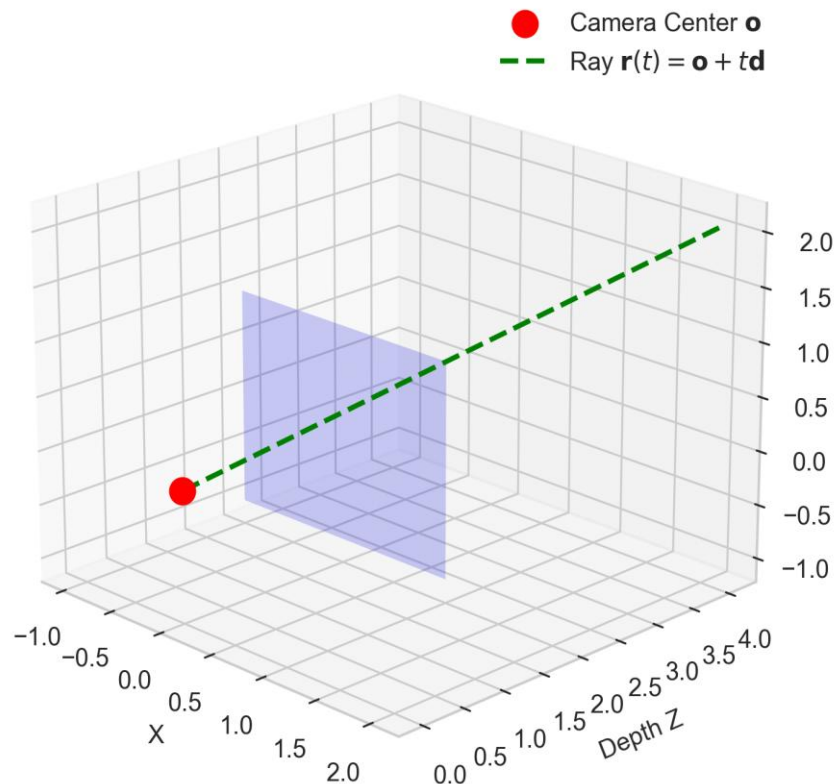
$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}, \quad t \in [t_n, t_f]$$

where $\mathbf{d} = \mathbf{R}^T \mathbf{K}^{-1}(u, v, 1)^T / \|\mathbf{K}^{-1}(u, v, 1)^T\|$

is the normalized direction.

Limitation of Classical SfM/MVS: Explicit geometry struggles with view-dependent effects and incomplete reconstruction.

Pinhole Camera Ray Casting



Shared Foundation — Volume Rendering Equation

Continuous Formulation: Integrates color along a ray through a participating medium.

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t) \sigma(\mathbf{r}(t)) \mathbf{c}(\mathbf{r}(t), \mathbf{d}) dt$$

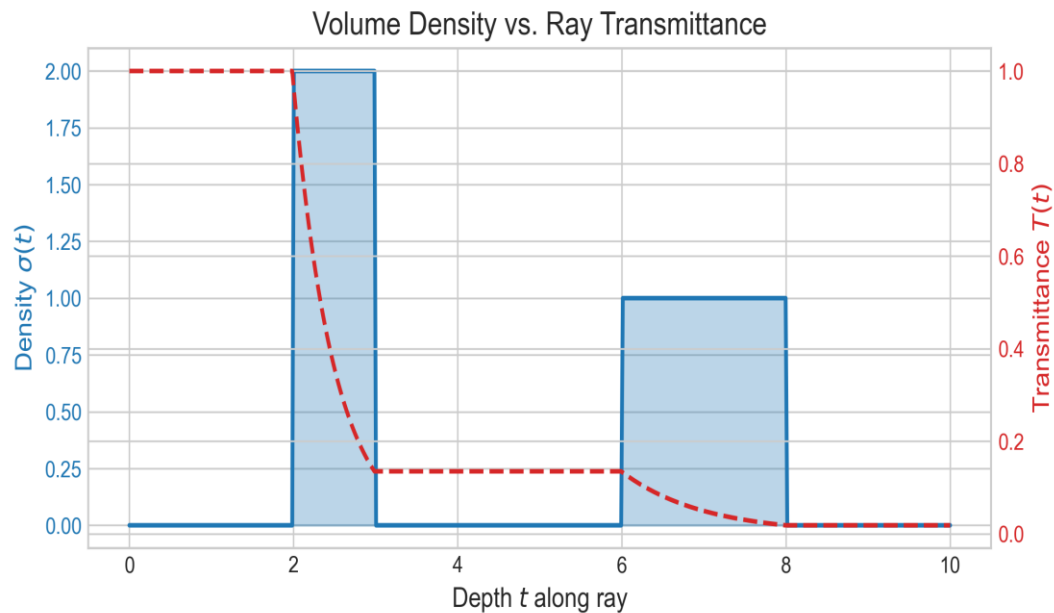
Transmittance $T(t)$:Probability rays reach depth t without being absorbed:

$$T(t) = \exp\left(-\int_{t_n}^t \sigma(\mathbf{r}(s)) ds\right)$$

Discrete Quadrature Approximation (Differentiable):

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N T_i \alpha_i \mathbf{c}_i, \quad \alpha_i = 1 - e^{-\sigma_i \delta_i},$$
$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$$

Geometric Insight: $T(t)\sigma(t)dt$ is the probability density that the ray terminates at exactly depth t .



Shared Foundation — Spherical Harmonics

Purpose: Represent view-dependent appearance (e.g., specularities) rigorously without deep networks.

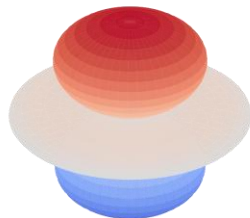
Function Approximation via Basis: Continuous color function of direction $\mathbf{d} \in \mathbb{S}^2$ decomposed into SH basis functions Y_l^m :

$$c(\mathbf{d}) = \sum_{l=0}^{l_{\max}} \sum_{m=-l}^l k_l^m Y_l^m(\mathbf{d})$$

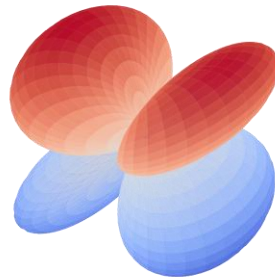
Capacity: At $l_{\max} = 3$) used in 3DGS), requires $(l_{\max} + 1)^2 = 16$ optimizable coefficients k_l^m per RGB channel.

Mathematical Elegance: Provides a smooth, orthogonal functional basis, inherently regularizing the view-dependent signals.

Y_2^0 Basis



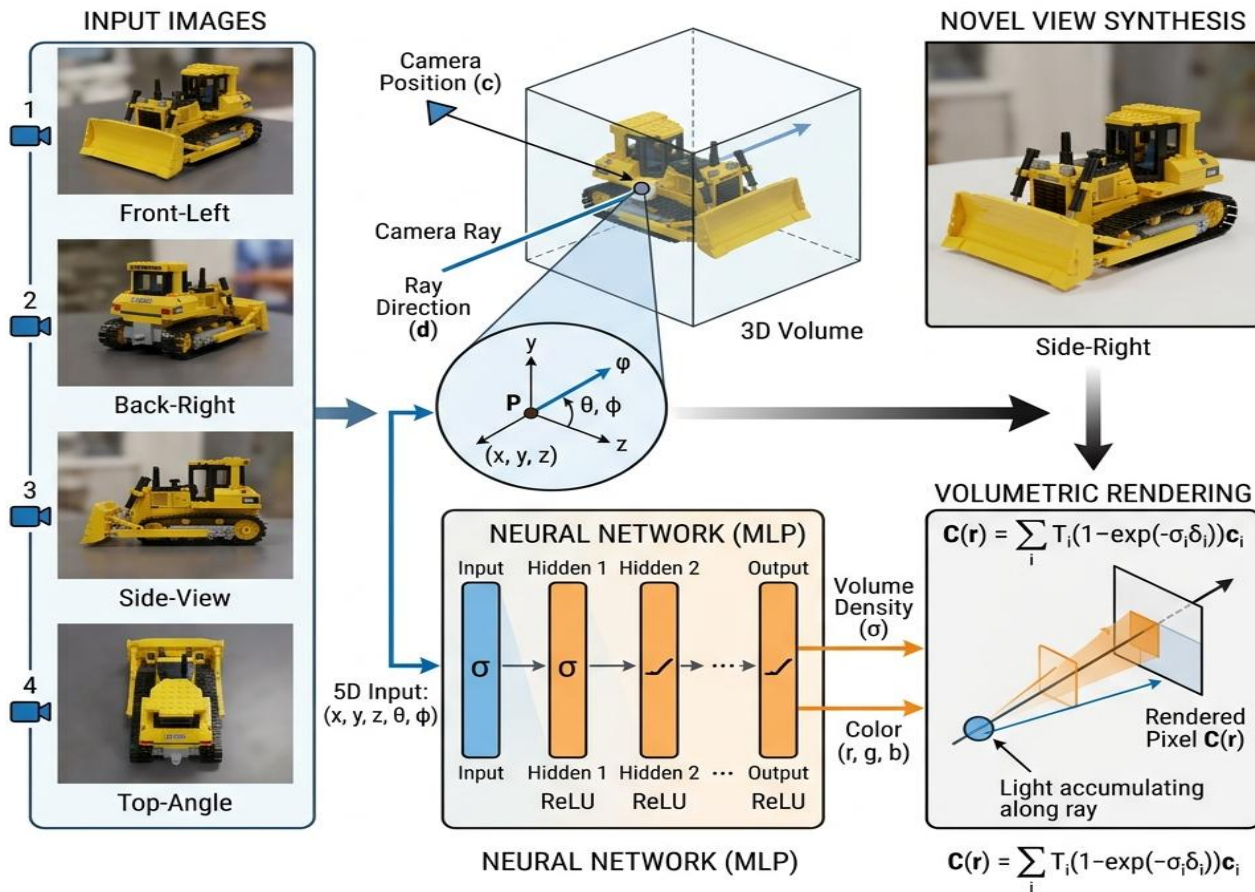
Y_2^1 Basis



Neural Radiance Fields (NeRF) — Core Logic

- **Implicit Representation:** Scene geometry and radiance encoded continuously via an MLP: $F_{\theta}: (\mathbf{x}, \mathbf{d}) \rightarrow (\mathbf{c}, \sigma)$.
- **Architecture:** 8 fully-connected layers (256 units, ReLU).
- **Multi-View Constraint:**
 - **Density** σ strictly depends only on position $\mathbf{x}$ (geometry is fixed across views).
 - **Radiance** $\mathbf{c}$ depends on $(\mathbf{x}, \mathbf{d})$ jointly, separated architecturally to enforce view-consistency.

Neural Radiance Fields



NeRF — Positional Encoding

Theorem Context (NTK Theory): Coordinate-based MLPs exhibit “spectral bias”, acting as strong low-pass filters that fail to capture high-frequency details.

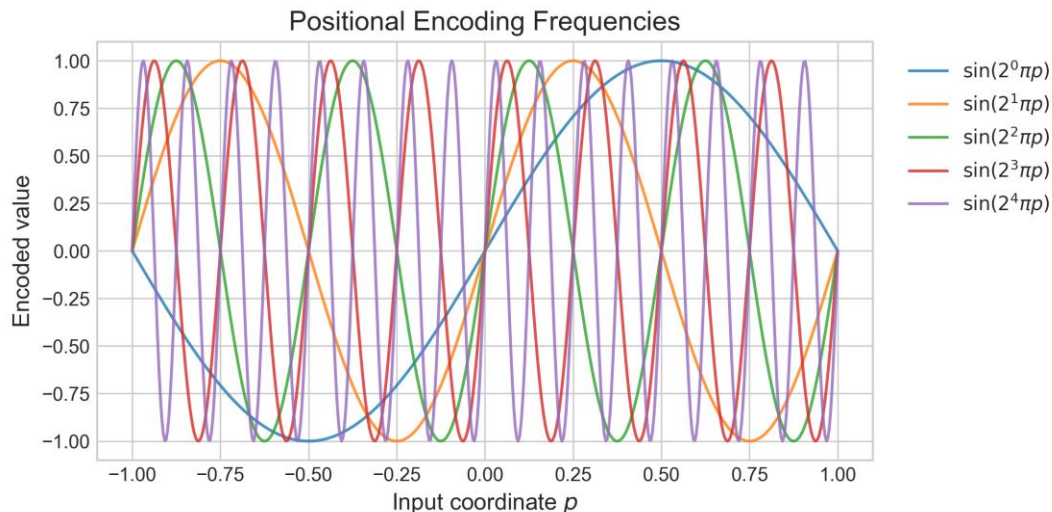
Solution: Lift input coordinates to a higher-dimensional space with varying frequencies:

$$\gamma(p) = (\sin(2^0\pi p), \cos(2^0\pi p), \dots, \sin(2^{L-1}\pi p), \cos(2^{L-1}\pi p))$$

Dimensions:

$L = 10$ for spatial mapping $\mathbf{x} \in \mathbb{R}^3 \rightarrow \mathbb{R}^{60}$;
 $L = 4$ for directional mapping $\mathbf{d} \in \mathbb{S}^2 \rightarrow \mathbb{R}^{24}$.

Insight: Approximates Random Fourier Features to allow arbitrary bandwidth kernel regression, saving NeRF from producing fully blurred imagery.



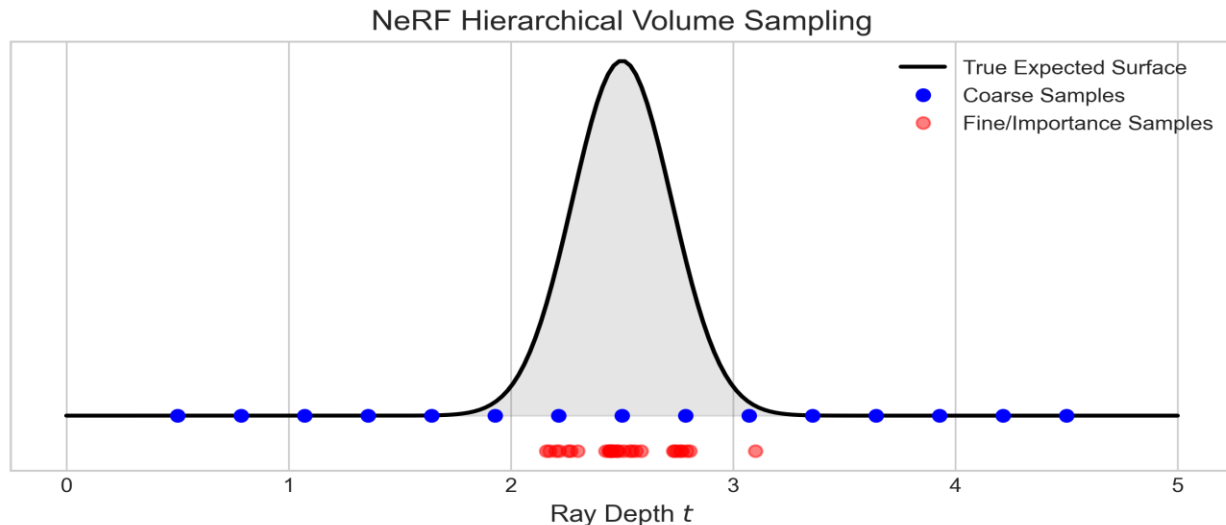
NeRF — Hierarchical Volume Sampling

Problem: Uniform integration heavily wastes evaluations in empty space.

Two-Network Importance Sampling:

1. **Coarse Distribution:** Compute unnormalized probabilities $w_i = T_i \alpha_i$ via N_c stratified samples.
2. **PDF Construction:** Create a piecewise-constant PDF $\hat{w}_i = w_i / \sum w_j$.
3. **Fine Distribution:** Draw N_f additional points via Inverse Transform Sampling.

Result: Computation optimally directed towards physical surfaces ($\sigma > 0$).



NeRF — Training Objective & Performance

- **Optimization:** Jointly optimizing Coarse and Fine networks by minimizing photometric L_2 error on rays $\mathcal{R}$:

$$\mathcal{L} = \sum_{\mathbf{r} \in \mathcal{R}} [\| \hat{C}_c(\mathbf{r}) - C_{gt}(\mathbf{r}) \|_2^2 + \| \hat{C}_f(\mathbf{r}) - C_{gt}(\mathbf{r}) \|_2^2]$$

- **Metrics:** Excellent PSNR/SSIM, but mathematically bounded by MLP inference scalability.
- **The Bottleneck:** $(N_c + N_f)$ MLP queries per ray $\rightarrow$ millions of queries per image $\rightarrow$ 30 seconds/frame render time.

3D Gaussian Splatting (3DGS) — Core Paradigm

- **Explicit Geometry:** Abandons MLPs for a collection of M point-based continuous primitives.

- **Anisotropic 3D Gaussian Primitive parameters**

$$\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \alpha_k, \mathbf{k}_k^{SH}\}:$$

Mean $\boldsymbol{\mu}_k$, Covariance $\boldsymbol{\Sigma}_k$, Opacity $\alpha_k \in [0,1]$, SH coefficients.

- **Volumetric Influence Function:**

$$G_k(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right)$$

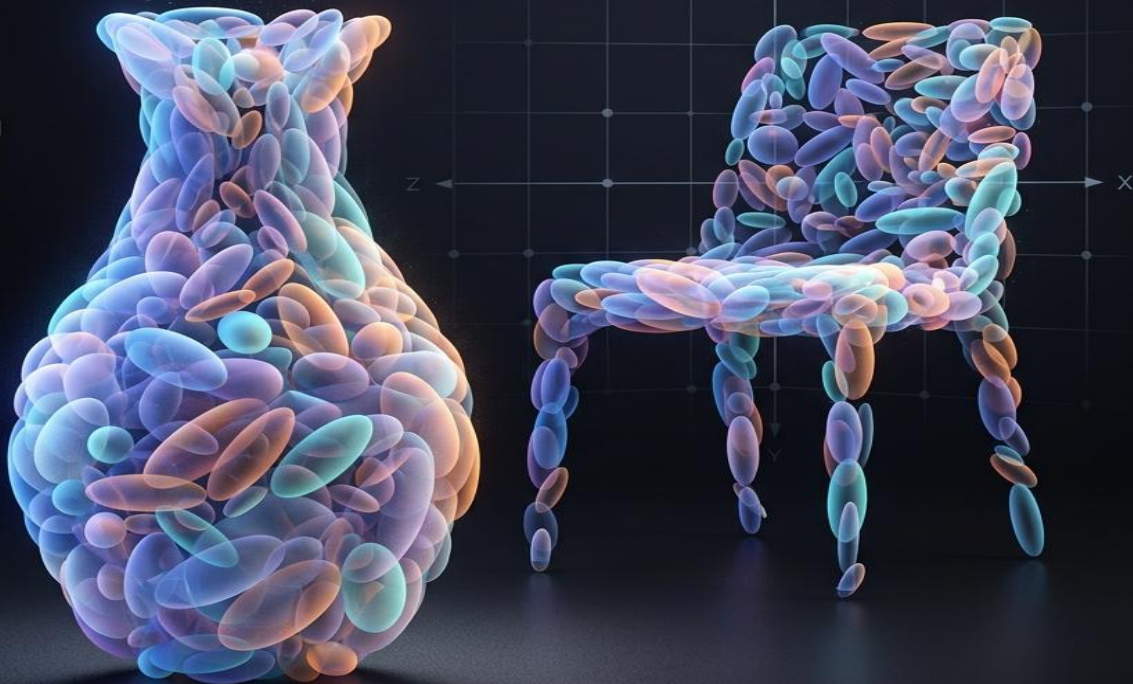
ANISOTROPIC GAUSSIAN PRIMITIVES

3D GAUSSIAN SPLATTING

ANISOTROPIC GAUSSIAN



$$\begin{aligned}\Sigma &= [3, 0, 1] \\ \mu &= [-0, -2, -3, 3] \\ \alpha &= 1.5 \\ \theta &= 20^\circ, \pm 0^\circ\end{aligned}$$



3DGS — Covariance Parameterization

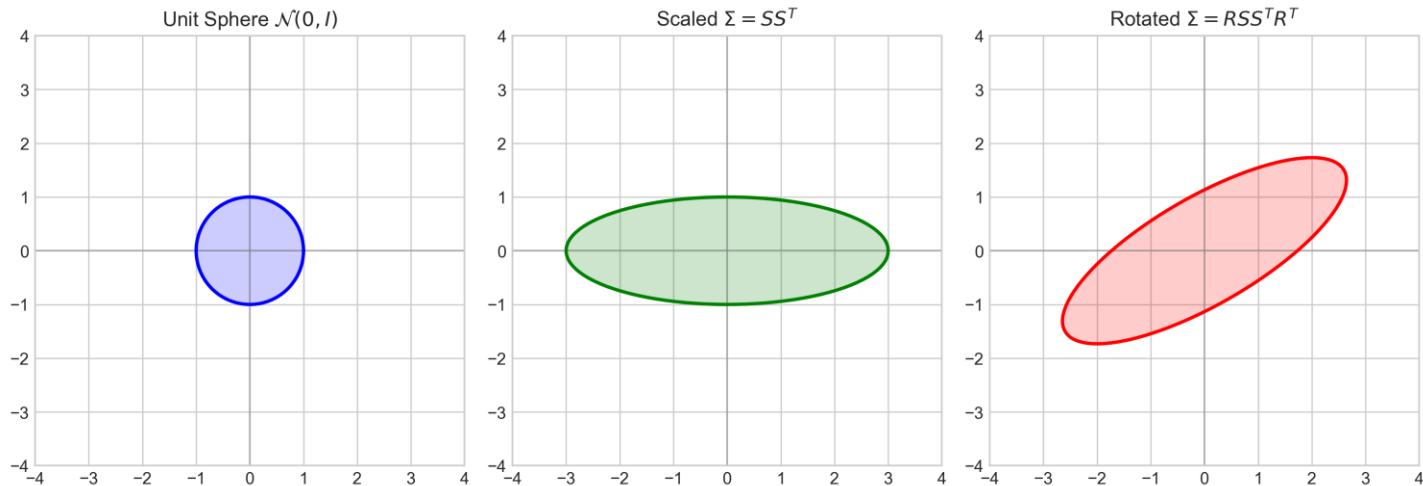
Challenge: Unconstrained gradient descent on Σ ruins positive semi-definiteness (PSD).

Cholesky-like Decomposition: Optimize instead via: $\Sigma = \mathbf{R}\mathbf{S}\mathbf{S}^T\mathbf{R}^T$

Rotation $\mathbf{R}$: Parameterized on the $SO(3)$ manifold via unit quaternion $\mathbf{q} \in \mathbb{S}^3$.

Scale $\mathbf{S}$: Diagonal matrix from exponential activation of $s \in \mathbb{R}^3$.

Degrees of Freedom Match: 3 DOFs from normalized quaternion + 3 from scale = exact 6 parameters for a valid 3×3 symmetric matrix.



3DGS — EWA Splatting (2D Projection)

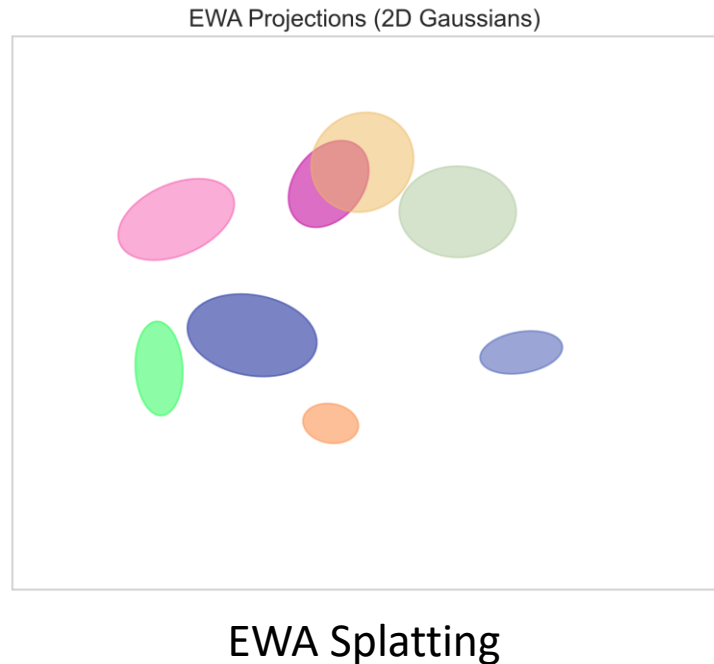
- **Goal:** Project arbitrary 3D ellipsoids accurately to the 2D image plane.
- **Elliptical Weighted Average (EWA) Map:**

$$\Sigma' = \mathbf{J}\mathbf{W}\Sigma\mathbf{W}^T\mathbf{J}^T$$

$\mathbf{W}$ is the affine world-to-view matrix.

$\mathbf{J}$ is the Jacobian of the projective transformation (handling the nonlinear perspective divide $1/z$).

- The final 2D covariance is simply the 2×2 upper-left block of Σ' .



3DGS — Alpha-Compositing & Differentiable Rasterizer

Front-to-Back accumulation math: Let 2D Gaussian influence be $G_i^{2D}(\mathbf{u})$. The pixel $\mathbf{u}$ color is:

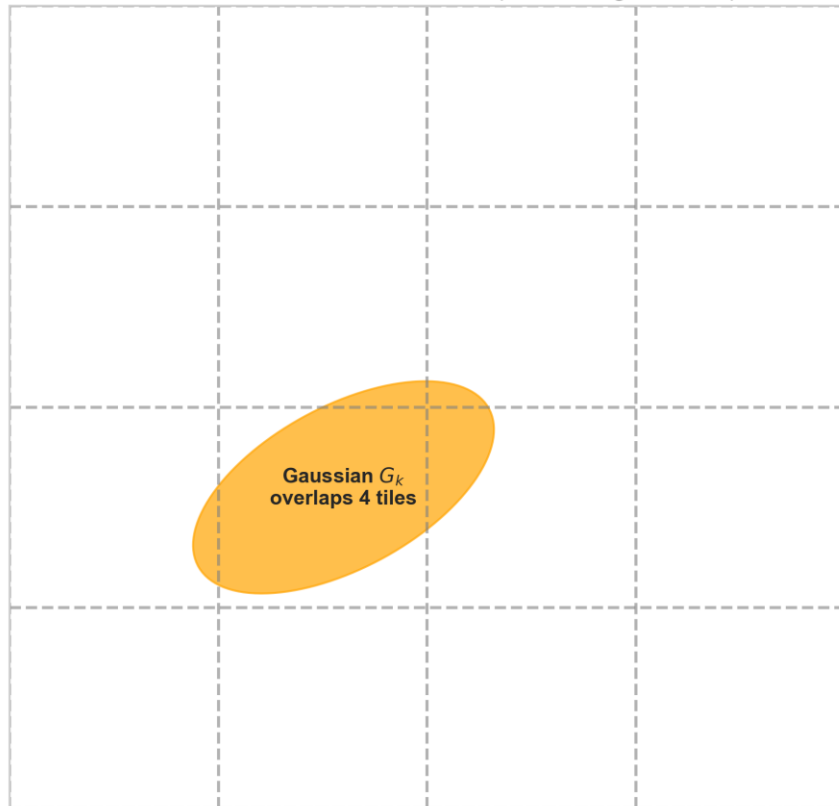
$$C(\mathbf{u}) = \sum_{i \in \mathcal{N}} \mathbf{c}_i \alpha_i G_i^{2D}(\mathbf{u}) \prod_{j=1}^{i-1} (1 - \alpha_j G_j^{2D}(\mathbf{u}))$$

Tile-Based CUDA Engine:

1. **Frustum Cull** → **Tile Matrix** (16x16 px).
2. **Fast GPU Radix Sort:** Sort depth keys sub-millisecond.
3. **Parallel Thread Accumulation:** Block loads sorted Gaussians to shared memory.

Enables ≥ 100 FPS fully-differentiable rasterization.

Tile-Based Rasterization Grid (16x16 logical tiles)



3DGS — Optimization & Adaptive Density Control

- **Rigorous Loss Construction:** Balances pixel-exactness with local structural perception (SSIM).

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_1 + \lambda \mathcal{L}_{\text{D-SSIM}}, \quad \lambda = 0.2$$

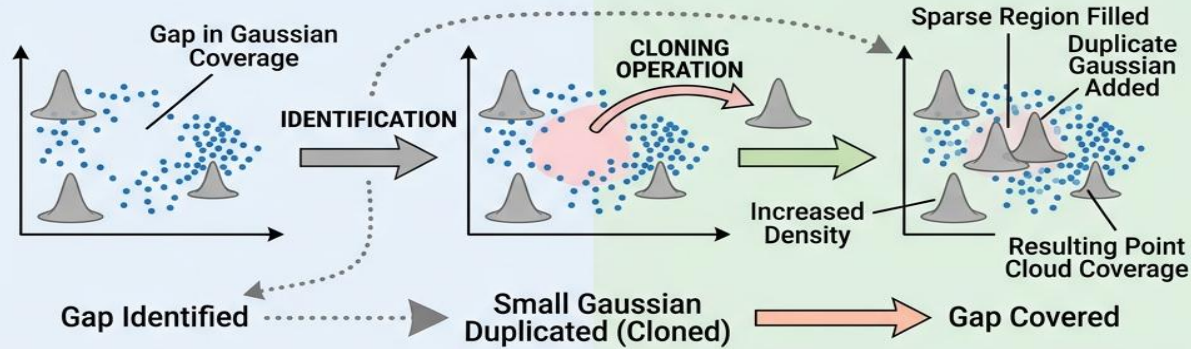
- **Adaptive Densification Topology (Every 100 iters):**

Targets gradients $> \tau_{\text{pos}} = 0.0002$.

- **Clone:** Duplicate & displace small Gaussians (resolves under-reconstruction).
- **Split:** Divide a large Gaussian into two smaller ones (scale division $\phi = 1.6$, resolves over-reconstruction).
- **Prune:** Cull transparent floaters dynamically ($\alpha < \epsilon_\alpha$).

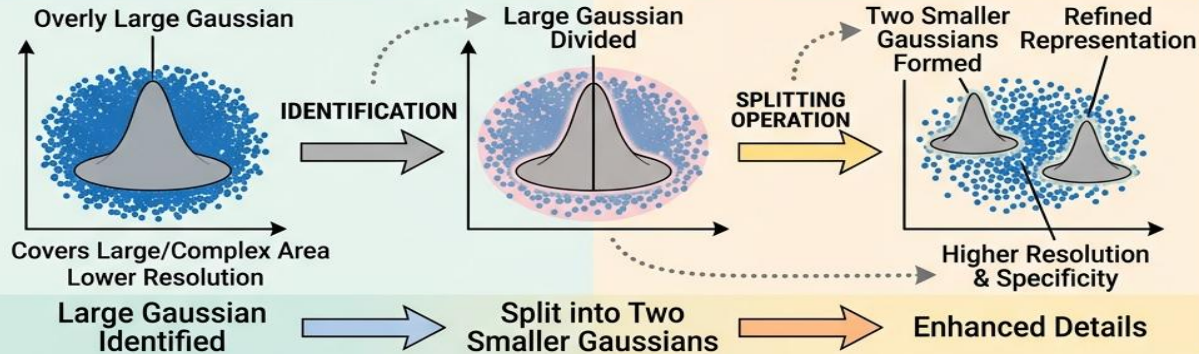
ADAPTIVE DENSITY CONTROL IN COMPUTER VISION

PART A: CLONING



Cloning: Identifies sparse areas in sparse regions and duplicates small Gaussians to increase coverage and density, filling gaps in representation.

PART B: SPLITTING



Splitting: Identifies large, complex Gaussians and divides them into two smaller, more focused Gaussians for higher resolution and accurate feature detail representation.

Identical Image Formation Physics

- **Compare Volume Rendering (NeRF):**

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N \left(\prod_{j<i} (1 - \alpha_j) \right) \alpha_i \mathbf{c}_i$$

- **With Alpha-Compositing (3DGS):**

$$C(\mathbf{u}) = \sum_{i \in \mathcal{N}} \left(\prod_{j<i} (1 - \alpha_j G_j^{2D}) \right) \alpha_i G_i^{2D} \mathbf{c}_i$$

- **Conclusion:** Both solutions solve analytically identical discretized integrals.
- **Core Pivot:** The breakthrough of real-time SOTA is fully algorithmic (GPU Sort + Rasterization), bypassing ray intersections while preserving physical light transport assumptions.

Ground Truth



NeRF



3D Gaussian Splatting



Error Map



Why is NeRF Blurry vs 3DGS Sharp?

1. **Spectral Bias of MLPs:** Even with coordinate encoding, MLPs act as continuous interpolators, applying an implicit “low-pass filter”.
2. **Volumetric Dispersion:** Ray marching integrates densities across intervals; minor errors in gradient descent “smear” boundaries into fog.
3. **Capacity Boundaries:** NeRF shares 1.2M network weights across the full volume. 3DGS employs dynamic allocation, generating millions of explicitly tiny Gaussian splats exactly where edge discontinuities demand it.

Connection to Course Topics (GMCV)

- **Structure-from-Motion (SfM):** Direct algorithmic predecessor; COLMAP point clouds form the strict initialization anchor for μ_k .
- **Projective Geometry:** Essential for deriving proper Jacobian matrices $\mathbf{J}$ for correct EWA mapping through perspective divides.
- **Optimization on Manifolds:** The use of $SO(3)$ and unit quaternions $\mathbb{S}^3$ to restrict geometry evolution explicitly maps strictly to GMCV rigid body theories.
- **Epipolar Geometry:** Replaced practically but enforced implicitly via multi-view reconstruction gradients.

Limitations & Open Problems

- **Representation Density:** 3DGS demands extensive memory for 1M-5M parameters per scene.
- **Surface Extraction:** Lacking implicit level-sets, extracting rigorous mesh topologies (like Marching Cubes) from Gaussian clouds remains mathematically ill-defined.
- **Artifacts:** Approximate depth sorting causes minor “popping” defects at extreme view alterations.
- **Mutual Dependency:** Both methodologies crash without absolute prior precision in classical camera calibrations.

Conclusion & Selected References

- **Domain Shift:** NVS has fully transitioned from computationally expensive, implicit field inference models toward intelligent, differentiable graphics-native rasterization primitives.
- **References:**
 - [1] Mildenhall et al. “NeRF: Representing Scenes as Neural Radiance Fields” (ECCV 2020)
 - [2] Kerbl et al. “3D Gaussian Splatting for Real-Time Radiance Field Rendering” (SIGGRAPH 2023)
 - [3] Zwicker et al. “EWA Volume Splatting” (IEEE Visualization 2001)
 - [4] Tancik et al. “Fourier Features Let Networks Learn High Frequency...” (NeurIPS 2020)